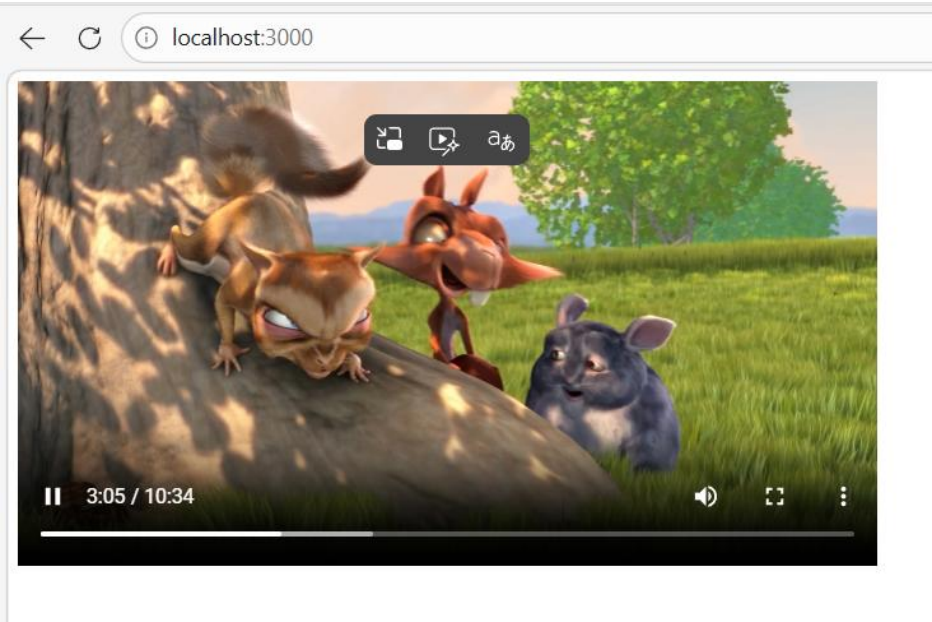


Demo screenshots:

Running docker compose:

```
$ docker-compose up --build
time="2026-07-05T09:11:14+02:00" level=warning msg="C:\\Users\\pm101\\OneDrive\\Desktop\\zeal\\video-streaming-analytics\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Building 3.1s (31/31) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 1:780s                                         0.0s
=> [processor internal] load build definition from Dockerfile            0.1s
=> => transferring dockerfile: 526B                                       0.0s
=> [client internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 86B                                         0.0s
=> [backend internal] load build definition from Dockerfile             0.1s
=> => transferring dockerfile: 217B                                         0.0s
=> [processor internal] load metadata for docker.io/apache/spark:4.0.0  0.9s
=> [backend internal] load metadata for docker.io/library/python:3.11   0.6s
=> [client internal] load metadata for docker.io/library/nginx:alpine  0.9s
=> [backend internal] load dockerignore                                 0.0s
=> => transferring context: 2B                                              0.0s
=> [backend 1/5] FROM docker.io/library/python:3.11sha256:7fca2f201542db9232b551db50cf4c4930d8baad691dbc973e1720cbf1b3756 0.1s
=> => resolve docker.io/library/python:3.11sha256:7fca2f201542db9232b551db50cf4c4930d8baad691dbc973e1720cbf1b3756 0.1s
=> [backend internal] load build context                                0.0s
=> => transferring context: 1.59kB                                         0.0s
=> CACHED [backend 2/5] WORKDIR /app                                    0.0s
=> CACHED [backend 3/5] COPY requirements.txt .                          0.0s
=> CACHED [backend 4/5] RUN pip install -r requirements.txt             0.0s
=> CACHED [backend 5/5] COPY                                             0.0s
=> [backend] exporting to image                                         0.4s
=> => exporting layers                                                    0.0s
=> => exporting manifest sha256:b09896f4e3692761366b31a1084b974f8ede1badd541c363e05302d99d422085 0.0s
=> => exporting config sha256:52aaf41b463fad1b040c27b1e51a9a10f51dd8a92d9eab1bc32d9b5f236d7da1 0.0s
=> => exporting attestation manifest sha256:f61d430ff23a1dca0a2e13a321bc001781ba70adfac8d6a75d112cd82bb11f 0.1s
=> => exporting manifest list sha256:a89334085d61cf8632a4be32c739c75ccdb7c3ca72c66c44aba1cc293 0.1s
=> => naming to docker.io/library/video-streaming-analytics-backend:latest 0.0s
=> => unpacking to docker.io/library/video-streaming-analytics-backend:latest 0.0s
=> [processor internal] load dockerignore                                 0.0s
=> => transferring context: 2B                                              0.0s
=> [client internal] load dockerignore                                  0.1s
=> => transferring context: 2B                                              0.0s
=> [processor 1/5] FROM docker.io/apache/spark:4.0.0sha256:a89782d90529a623fc4471cdddb0f9c32d6eed10a81b256a80207b23c31d4f00 0.2s
```

Starting the video:



Logs for the events being processed by the pipeline:

```
backend-1 | Received: {'event_id': '3kult', 'timestamp': 1783235574302, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 196.118, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:36284 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': '10p3w', 'timestamp': 1783235579307, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 201.123, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:53194 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'x2mq4', 'timestamp': 1783235584299, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 206.114, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:53194 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'x7yy4', 'timestamp': 1783235589303, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 211.119, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:53194 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'c3fdm', 'timestamp': 1783235594308, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 216.124, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:59358 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': '1i0wtm', 'timestamp': 1783235599304, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 221.12, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:59358 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'nrkjd', 'timestamp': 1783235604302, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 226.118, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:59358 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': '0lba09', 'timestamp': 1783235609301, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 231.113, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:59358 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'abdjmrp', 'timestamp': 1783235614311, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'HEARTBEAT', 'video_position': 236.124, 'bitrate': 836}
backend-1 | INFO: 172.18.0.1:49438 - "POST /telemetry HTTP/1.1" 200 OK
backend-1 | Received: {'event_id': 'k804ro', 'timestamp': 1783235618242, 'viewer_id': 'viewer_165', 'session_id': 'btvcli', 'event_type': 'QUALITY_CHANGE', 'video_position': 240.053, 'level': 3, 'bitrate': 836}
```

Quering data in clickhouse:

Telemetry data:

http://localhost:8123

default

select * from default.telemetry_events;

Run

(Ctrl/Cmd+Enter, +Shift to run all)

24 rows in result, 0.01 sec.

100.0%, Read 24 rows, 1.46 KB

#	event_id	timestamp	viewer_id	session_id	event_type	video_position	bitrate	processed_at
1	lv4c93	1783237760393	viewer_197	s0ecum	QUALITY_CHANGE	268.088	6222	2026-07-05 07:49:30
2	7lfbj	1783237760394	viewer_197	s0ecum	BUFFER_END	268.088	6222	2026-07-05 07:49:30
3	eejagl	1783237762431	viewer_197	s0ecum	QUALITY_CHANGE	270.056	836	2026-07-05 07:49:30
4	bzjn3t	1783237764003	viewer_197	s0ecum	BUFFER_START	451.728	836	2026-07-05 07:49:30
5	x4yzl	1783237764069	viewer_197	s0ecum	HEARTBEAT	451.728	836	2026-07-05 07:49:30
6	x7nl9s	1783237766782	viewer_197	s0ecum	QUALITY_CHANGE	451.728	2149	2026-07-05 07:49:30
7	96lo5q	1783237766782	viewer_197	s0ecum	BUFFER_END	451.728	2149	2026-07-05 07:49:30
8	lg6j67	1783237769067	viewer_197	s0ecum	HEARTBEAT	453.403	2149	2026-07-05 07:49:30
9	hmgcnp	1783237734420	viewer_165	btvc1i	HEARTBEAT	7.813	2149	2026-07-05 07:49:00
10	6qwk5	1783237739414	viewer_165	btvc1i	HEARTBEAT	12.807	2149	2026-07-05 07:49:00
11	24dvmh	1783237722567	viewer_165	btvc1i	HEARTBEAT	307	836	2026-07-05 07:48:50
12	6kkoka	1783237724421	viewer_165	btvc1i	HEARTBEAT	307	836	2026-07-05 07:48:50
13	clzn8c	1783237725777	viewer_165	btvc1i	BUFFER_START	0	836	2026-07-05 07:48:50
14	f0s7ig	1783237726346	viewer_165	btvc1i	BUFFER_END	0	836	2026-07-05 07:48:50
15	syp5f	1783237726597	viewer_165	btvc1i	BUFFER_START	0.1	836	2026-07-05 07:48:50

Buffer duration events:

http://localhost:8123

default

select * from default.telemetry_events;

select * from default.buffer_durations;

select * from buffer_timeouts;

Run selected

Run all

(Ctrl/Cmd+Enter, +Shift to run all)

4 rows in result, 0.00 sec.

100.0%, Read 4 rows, 184.00 B

#	viewer_id	session_id	buffer_start_ts	buffer_end_ts	duration_ms	calculated_at
1	viewer_165	btvc1i	1783237725777	1783237726346	569	2026-07-05 07:48:52
2	viewer_165	btvc1i	1783237726597	1783237726602	5	2026-07-05 07:48:52
3	viewer_197	s0ecum	1783237756848	1783237760394	3546	2026-07-05 07:49:31
4	viewer_197	s0ecum	1783237764003	1783237766782	2779	2026-07-05 07:49:31

Grafana dashboard:

